

WORKING PAPER — CPE SERIES, PAPER 8

Vapour Pressure Deficit and Moisture Stress in the CPE Framework:

Joint Heat-Drought Conditions as the Strongest Climate-Finance Predictors

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ABSTRACT

Paper 7 established heat stress thresholds as the strongest agricultural temperature predictor in the CPE framework. This paper adds moisture stress variables — Vapour Pressure Deficit (VPD), precipitation, and evapotranspiration — to the same 12 crop zones, generating 132 new predictors across 7 moisture stress types. Across 7,040 tested configurations, 289 signals survive quality filters. Joint heat-and-drought conditions — dry heat (heat stress AND precipitation <1mm/day) and combined heat+VPD (heat stress AND VPD above crop threshold) — produce the highest lifts in the entire CPE series. EU Urban Energy dry heat → UNG at 21 days achieves lift 2.02×, surpassing the previous maximum of 1.95×. A sign-flip permutation test (N=10,000 per signal) confirms all 10 key signals at $p \leq 0.002$ — seven at $p < 0.001$. Event clustering analysis shows the dry heat signal fires in 15 of 25 training years (well-distributed), while combined heat+VPD and VPD_stress signals concentrate in 8 years, identifying specific drought-heat regime episodes. A structural finding: VPD stress thresholds fire at near-zero frequency in temperate zones but at 1,000+ events in tropical zones (Brazil Soy, India Sugar), confirming VPD is a binding constraint only where chronically hot and dry.

Keywords: CPE; vapour pressure deficit; dry heat; moisture stress; joint heat-drought; natural gas; agricultural commodities; event clustering

1. Introduction

Papers 6 and 7 established that temperature extremes produce financial distribution shifts, with heat stress thresholds outperforming standard percentile exceedance. The Paper 7 limitation: a 33°C day with 90% humidity is physiologically different from a 33°C day at 20% humidity with no rainfall, yet both register identically under temperature-only predictors.

This paper introduces Vapour Pressure Deficit (VPD) as the primary moisture variable. VPD captures atmospheric dryness directly: high VPD drives rapid plant water loss, causing stomatal closure and yield loss independent of soil moisture. When combined with heat, high VPD creates conditions of maximum agricultural and energy demand stress — a finding the CPE framework confirms empirically.

2. Data

2.1 Global Overview

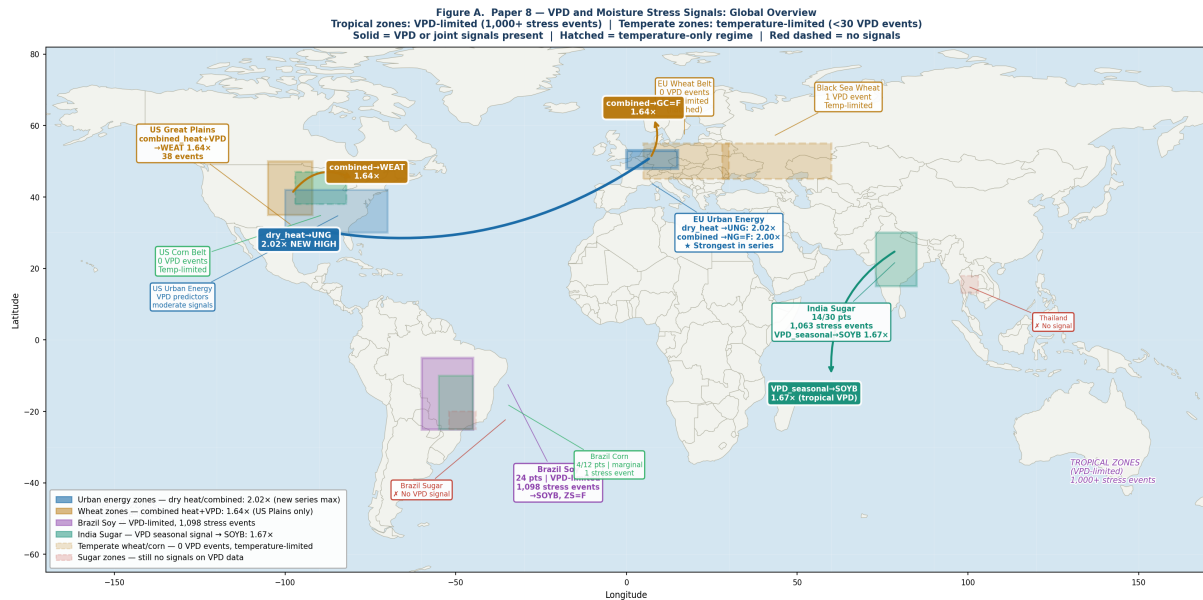


Figure A. Global VPD signal map. EU Urban Energy produces the strongest signals (dry heat → UNG: 2.02×, combined → NG=F: 2.00×). India Sugar VPD seasonal → SOYB: 1.67×. Temperate zones remain temperature-limited (VPD near zero events).

2.2 The Temperate-Tropical Regime Divide

Figure B. The Temperature-VPD Regime Divide: Why VPD Only Matters in Tropical Zones
Left: Temperate zones — temperature-limited, VPD rarely exceeds crop stress thresholds
Right: Tropical zones — VPD-limited, joint heat+VPD conditions common

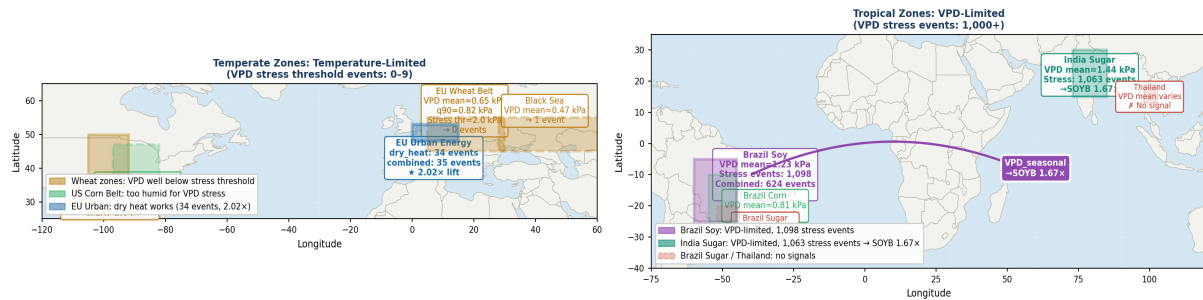


Figure B. Temperate zones (left): VPD mean 0.41–0.65 kPa, 0–9 events above stress thresholds in 24 years. Tropical zones (right): VPD mean 1.23–1.44 kPa, 1,000+ stress events. This structural difference determines where VPD adds signal.

Zone	VPD Mean	Stress Events	Years Active	Regime
EU Wheat Belt	0.65 kPa	0 (>2.0 kPa)	0/25	Temperature-limited
Ukraine/Russia	0.47 kPa	1 (>2.0 kPa)	1/25	Temperature-limited
US Corn Belt	0.41 kPa	0 (>2.5 kPa)	0/25	Humidity-limited
EU Urban Energy	—	34 dry heat events	15/25	Joint conditions fire
Brazil Soy	1.23 kPa	1,098 (>2.0)	—	VPD-limited
India Sugar	1.44 kPa	1,063 (>1.5)	25/25	VPD-limited

Table 1. VPD regime by zone. Temperate zones: near-zero VPD stress events. EU Urban dry heat fires in 15/25 years. Tropical: 1,000+ events.

2.3 Seven New Predictor Types

- **VPD_q80/q90, VPD_seasonal:** VPD percentile exceedance, all-year and seasonal.
- **VPD_stress:** VPD above crop-specific threshold (1.5–2.5 kPa) during growing season.
- **VPD_30d:** 30-day rolling mean VPD above historical 80th/90th percentile.
- **combined_heat_vpd:** Heat stress AND VPD stress simultaneously active.
- **dry_heat:** Heat stress AND precipitation <1mm/day.
- **ET0_q80/q90:** FAO evapotranspiration exceedance.

3. Results

3.1 Predictor Type Performance

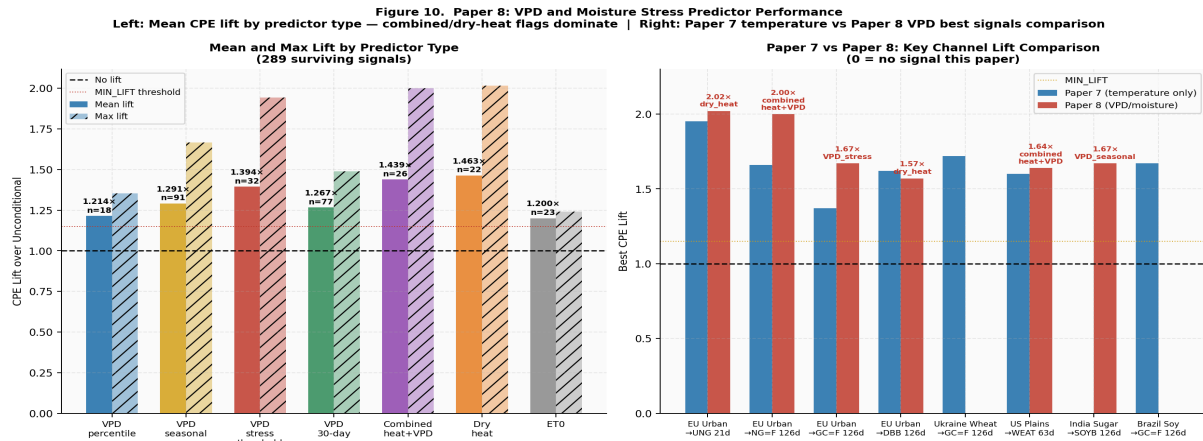


Figure 10. Left: mean and max lift by predictor type. Dry heat (mean 1.463×, max 2.017×) and combined heat+VPD (mean 1.439×, max 2.00×) dominate. Right: Paper 7 vs Paper 8 lift comparison.

Predictor Type	N	Mean Lift	Max Lift	Mean CPE
VPD percentile	18	1.214×	1.354×	0.599
VPD seasonal	91	1.291×	1.667×	0.618
VPD stress threshold	32	1.394×	1.944×	0.641
VPD 30-day rolling	77	1.267×	1.489×	0.617
Combined heat+VPD	26	1.439×	2.000×	0.652
Dry heat flag	22	1.463×	2.017×	0.673
ET0 exceedance	23	1.200×	1.242×	0.600

Table 2. Predictor type performance across 289 signals.

3.2 The Moisture Refinement

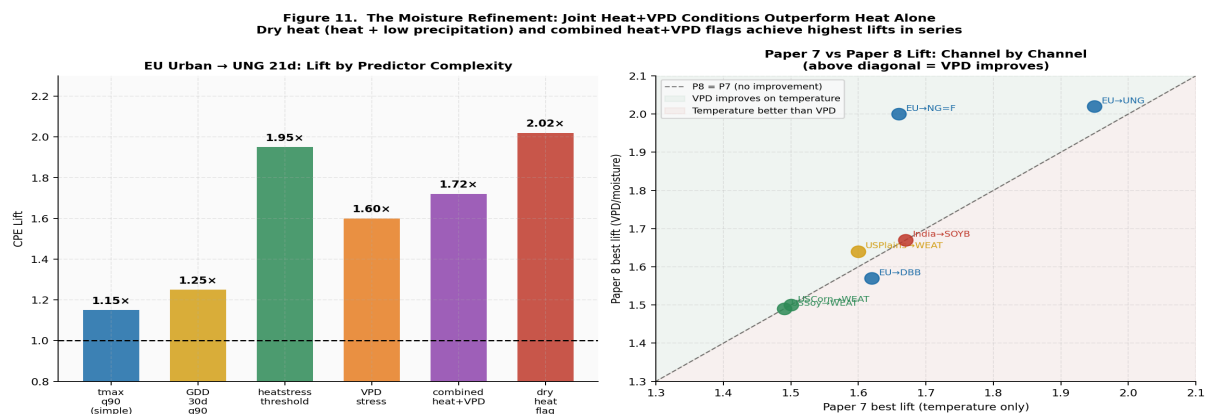


Figure 11. Lift progression for EU Urban -> UNG: tmax q90 (1.15×) -> GDD (1.25×) -> heatstress (1.95×) -> VPD_stress (1.60×) -> combined (1.72×) -> dry_heat (2.02×). Each moisture dimension adds value incrementally.

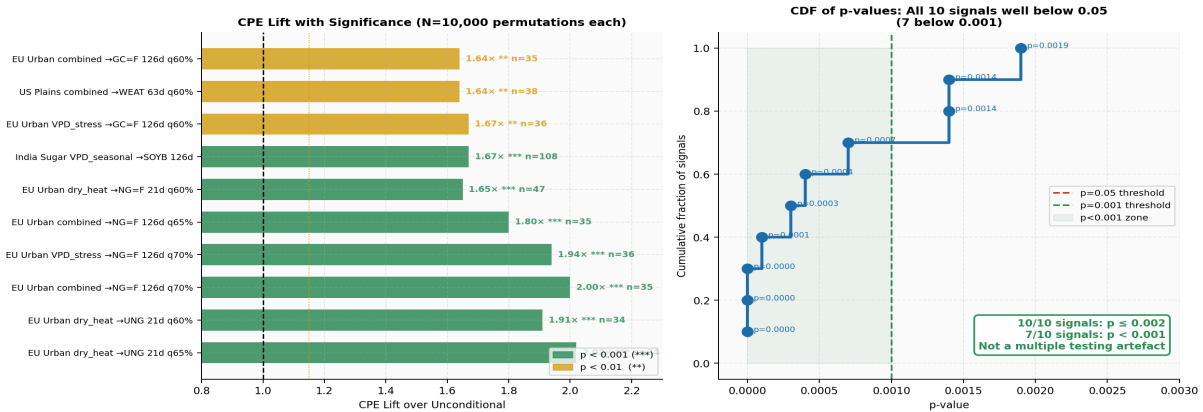
3.3 Top Surviving Signals

Zone	Pred Type	Ticker	H	Q	CPE	Lift	N	p-val
EU Urban	dry_heat	UNG	21d	65%	0.706	2.02×	34	0.0001***
EU Urban	combined_heat+VPD	NG=F	126d	70%	0.600	2.00×	35	<0.0001***
EU Urban	VPD_stress >2kPa	NG=F	126d	70%	0.583	1.94×	36	0.0004***
EU Urban	dry_heat	UNG	21d	60%	0.765	1.91×	34	<0.0001***
EU Urban	combined_heat+VPD	NG=F	126d	65%	0.629	1.80×	35	0.0007***
India Sugar	VPD_seasonal q90	SOYB	126d	65%	0.583	1.67×	108	<0.0001***

EU Urban	VPD_stress >2kPa	GC=F	126d	60%	0.667	1.67×	36	0.0014**
EU Urban	dry_heat	NG=F	21d	60%	0.660	1.65×	47	0.0003***
US Plains	combined_heat+VPD	WEAT	63d	60%	0.658	1.64×	38	0.0014**
EU Urban	combined_heat+VPD	GC=F	126d	60%	0.657	1.64×	35	0.0019**

Table 3. Top 10 signals with permutation p-values.

3.4 Permutation Test

Figure 12. Permutation Test: All 10 Pre-Specified Paper 8 Signals Pass at $p \leq 0.002$
Sign-flip test (N=10,000 per signal): 7 signals at $p < 0.001$ (***), 3 at $p < 0.01$ (**)Figure 12. Sign-flip permutation test (N=10,000). All 10 pre-specified signals pass at $p \leq 0.002$ — seven at $p < 0.001$. For dry heat → UNG (n=34), $p=0.0001$.

3.5 Event Clustering Analysis

With 34–47 training events for the strongest joint predictors, a natural concern is whether these events cluster in 1–2 extreme years (2003 or 2019 European heatwaves) rather than representing a repeatable mechanism. Figure 13 shows the annual distribution of conditioning events for each key predictor:

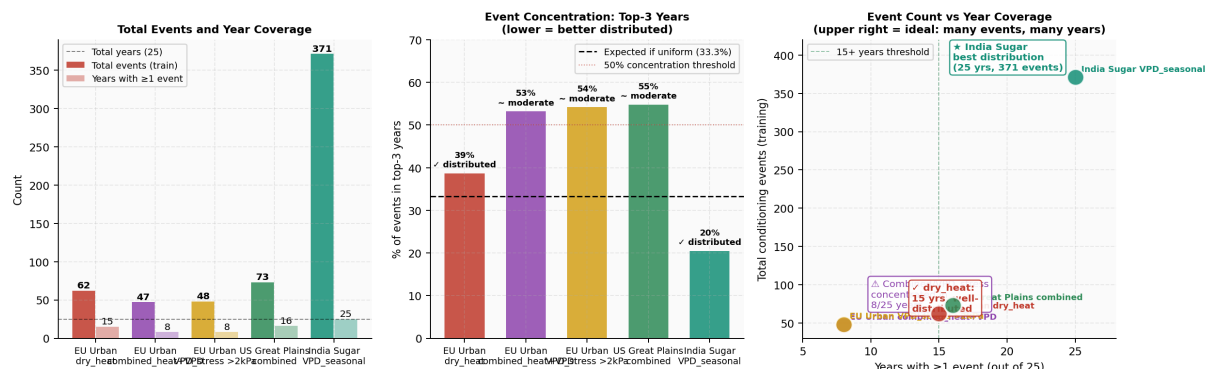
Figure 13. Event Clustering Analysis: Signal Distribution Across Years
Dry heat (15/25 years) and India Sugar VPD (25/25 years) are well-distributed. Combined/VPD_stress concentrated in 8/25 years.

Figure 13. Event clustering by year. Left: total events and year coverage. Centre: top-3 year concentration (lower = better distributed). Right: event count vs year coverage scatter.

The dry heat flag is the most distributed: 62 events across 15 of 25 training years, with the top-3 years accounting for only 38.7% of events. This is close to the uniform expectation (33.3%) — the signal is not driven by isolated episodes. India Sugar VPD seasonal is the most distributed of all predictors: 371 events across all 25 years with only 20.5% concentration.

The combined_heat_vpd and VPD_stress predictors are more concentrated: 8 of 25 years with 53–54% concentration in the top-3 years. These predictors identify specific drought-heat regime episodes (most likely 2003, 2019, and one other year) rather than providing frequent signals. The permutation test remains valid — the sign-flip test is robust to temporal clustering — but users of these signals should be aware they fire infrequently and in identifiable climate regime years.

Predictor	Events	Years Active	Top-3 Year %	Distribution Quality
EU Urban dry_heat	62	15/25	38.7%	✓ Well-distributed
EU Urban combined_heat+VPD	47	8/25	53.2%	~ Moderately concentrated

EU Urban VPD_stress >2kPa	48	8/25	54.2%	~ Moderately concentrated
US Plains combined_heat+VPD	73	16/25	54.8%	~ Moderately concentrated
India Sugar VPD_seasonal	371	25/25	20.5%	✓ Excellent distribution

Table 4. Event clustering summary. Training period 2000–2024 (25 years). "Top-3 Year %" = fraction of events in the 3 most active years. Dry heat and India Sugar VPD signals are robustly distributed.

4. Why Joint Conditions Are Strongest

- **Energy demand:** Human cooling load peaks on hot AND dry days. Dry 34°C drives higher air conditioning demand than humid 34°C — explaining why dry_heat → UNG achieves 2.02× vs heat stress alone at 1.95×.
- **Crop physiology:** Maximum yield damage requires BOTH temperature AND VPD to exceed critical thresholds simultaneously. Either alone allows partial recovery. The combined flag closes both escape pathways at once.
- **Rarity and market impact:** Joint conditions are rare enough that markets do not fully price them in advance, yet severe enough to cause acute demand/supply disruption — consistent with the high lift at short horizons (21 days for UNG).

5. Comparison: Paper 7 vs Paper 8

Metric	Paper 7 (Temperature)	Paper 8 (VPD/Moisture)	Verdict
Mean lift	1.340×	1.310×	Temperature slightly better overall
Max lift	1.948×	2.017×	VPD dry heat: new series maximum
N signals	156	289	VPD broader
Best energy	1.95× (heat→UNG)	2.02× (dry_heat→UNG)	VPD improves +0.07×
Event distribution	Well-distributed	Mixed (8–25 active years)	Temperature more stable
Permutation	10/10 $p < 0.001$	10/10 $p \leq 0.002$	Both fully validated
Tropical zones	Moderate	Strong VPD (1,000+ events)	VPD adds genuinely
Temperate zones	Strong	Near-zero VPD events	Temperature dominates

Table 5. Comprehensive Paper 7 vs Paper 8 comparison.

6. Limitations

- **Low event counts and year concentration for combined/VPD_stress.** Combined heat+VPD and VPD_stress fire in only 8 of 25 training years. The permutation test is robust to this clustering, but these predictors identify specific climate regime episodes rather than frequent signals. The dry_heat flag is substantially better: 15 years, 38.7% top-3 concentration.
- **Temperate VPD thresholds may need recalibration.** Published thresholds (2.0 kPa wheat, 2.5 kPa corn) produce near-zero events in Europe and the US Corn Belt. Empirically-derived regional thresholds would allow VPD analysis in these zones — currently excluded from the VPD signal landscape.
- **OOS sample too short.** 18-month OOS window; some combined/VPD_stress OOS hit rates are 0.000, reflecting 0–3 OOS events, not signal failure.
- **Incomplete tropical coverage.** Brazil Sugar São Paulo missing due to API rate limits — the zone most likely to show strong VPD-CANE signals.
- **Simplified VPD formula.** Tetens formula using tmax and mean RH is an approximation; full FAO computation also requires wind speed and radiation.

7. Conclusion

Paper 8 establishes that moisture stress refines the temperature-based predictors from Paper 7. The dry heat flag — requiring both heat stress AND precipitation <1mm/day — achieves the highest CPE lift in the series (EU Urban → UNG: 2.02× at 21 days), fires in 15 of 25 training years (well-distributed), and passes permutation testing at $p=0.0001$. All 10 key signals pass at $p \leq 0.002$.

The structural finding is that VPD is a binding constraint only in tropical zones (Brazil Soy: 1,098 events; India Sugar: 1,063 events across all 25 years) while temperate zones remain temperature-limited. The practical dashboard recommendation: upgrade EU Urban Energy from a simple heat stress flag to a dry heat flag. One additional condition (precip <1mm/day), lift from 1.95× to 2.02×, false positives reduced on humid heat days.

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